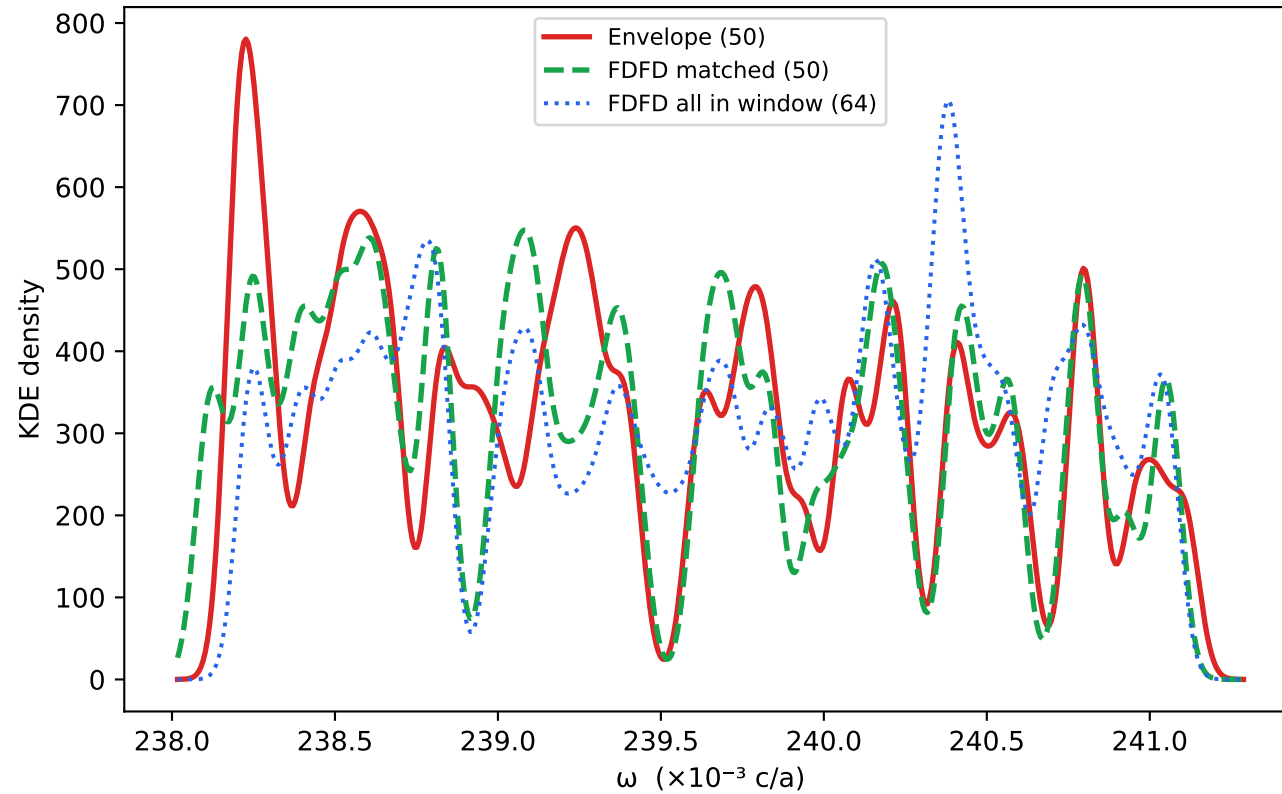
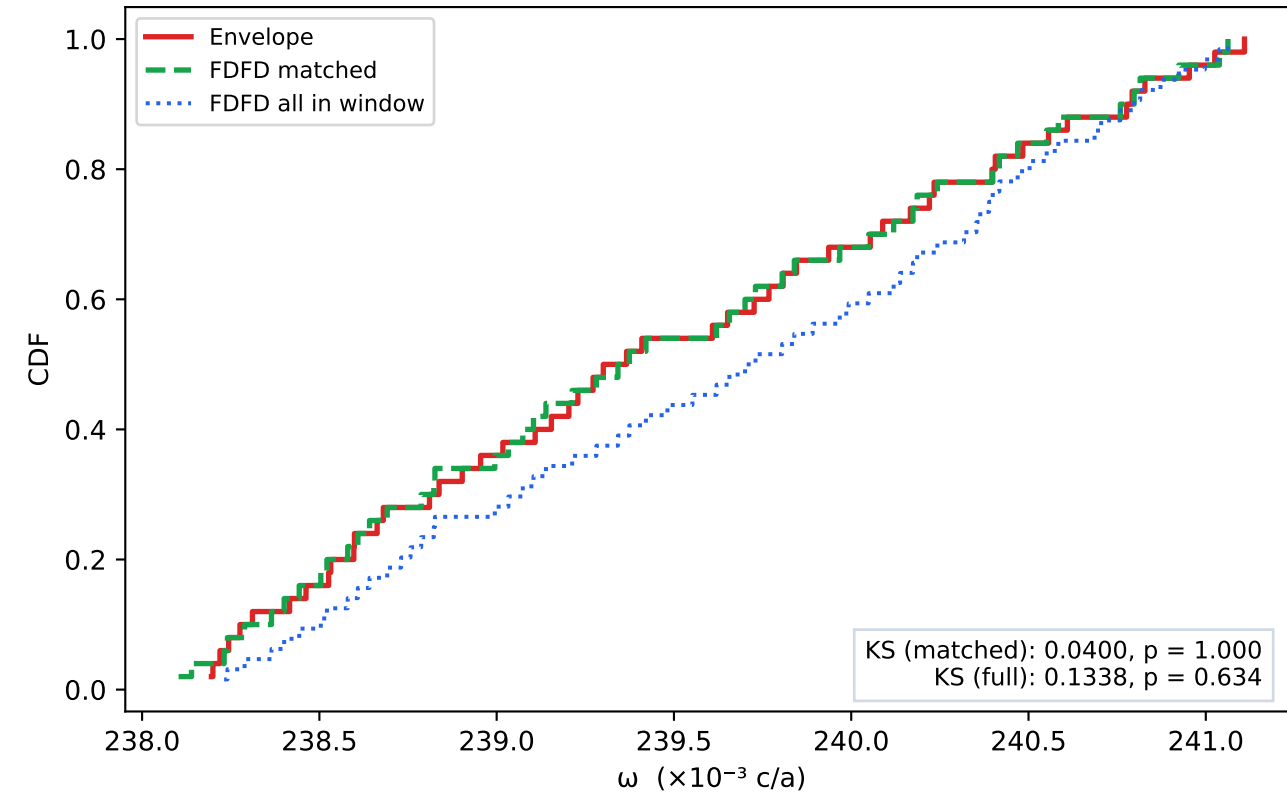


Spectral Structure Comparison | $\theta = 1.12^\circ$ | EA (2-band, 33K DOF) vs FDFD (4.2M DOF, res=40)

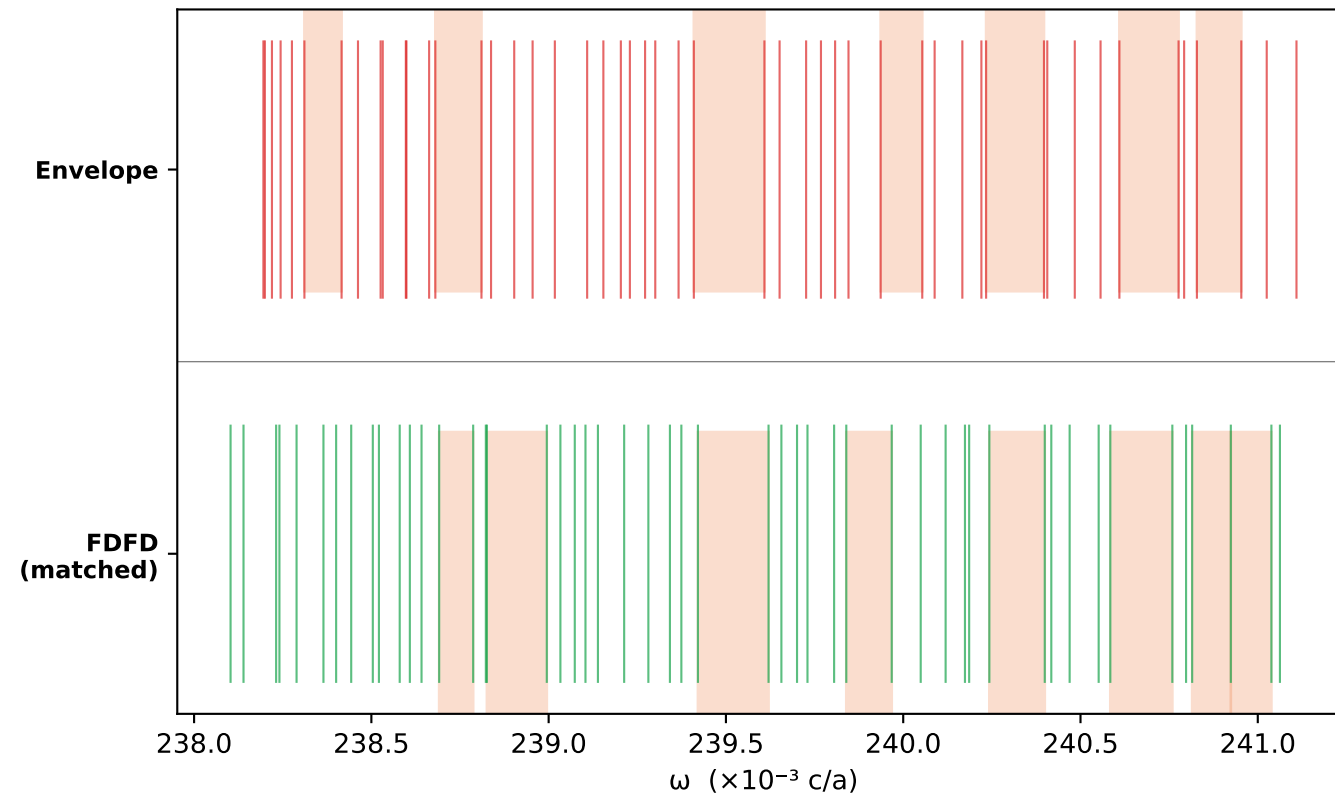
(a) Density of States (KDE)



(b) Cumulative Distribution Function



(c) Gap structure (7 EA gaps, 8 FDFD gaps)



SPECTRAL STRUCTURE SUMMARY

Bandwidth:

EA: 2.9137×10^{-3} c/a
FDFD: 2.9596×10^{-3} c/a
Ratio: 0.9845

Spectral center:

EA: 0.239471
FDFD: 0.239462
 Δ : $9.11e-06$

KS test (EA vs FDFD matched):

D = 0.0400, p = 1.000
→ Cannot reject same distribution

Relative error $|\Delta\omega|/\omega$:

mean = $9.69e-05$
max = $5.44e-04$

FDFD discretization error:

63×10^{-6} (res=16→20 drift)
EA-FDFD residual: 23×10^{-6}
→ FDFD grid error dominates